Land-Sea Target Detection and Recognition in SAR Image Based on Non-Local Channel Attention Network

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***Abstract*— Synthetic aperture radar (SAR) target recognition is essential for SAR image interpretation. It has been widely used in national defense and national economy. At present, the SAR image detection and recognition methods based on convolutional neural network (CNN) have problems such as insufficient extraction of the feature information of SAR image targets, false targets caused by the interference of complex backgrounds, and low detection performance. The main reason is that the feature extraction of CNN is a local operation in space and time, which ignores the correlation between pixels and regions and the dependencies between channels in SAR images. In this article, a non-local channel attention network (NLCANet) SAR image target recognition method is proposed based on the GoogLeNet structure combined with asymmetric pyramid non-local block (APNB) and squeeze-and-excitation block (SEB). APNB is added to the GoogLeNet framework to capture more context information and enhance the correlation between pixels and regions. SEB is added to the Inception structure to become Inception-SEB (ISEB), through which channel dependencies based on the fusion of different scale features can be obtained. The experimental results based on the moving and stationary target acquisition and recognition (MSTAR) dataset and the SAR ship detection dataset (SSDD) show that the proposed method improves the detection ability of targets in complex backgrounds and achieves better land-sea target recognition performance.**

***Index Terms*— Asymmetric pyramid non-local block (APNB), GoogLeNet, squeeze-and-excitation block (SEB), synthetic aperture radar (SAR) image, target recognition.**

NOMENCLATURE

List of Notations:

x	Feature map obtained by the traditional convolutions of the input synthetic aperture radar (SAR) image.
φ, θ, g	Input feature map is linearly mapped to obtain three features.
g_{p}	Feature g_{p} obtained by g through the spatial pyramid pooling (SPP) model.
W_{g}, W_θ, W_φ	It is represented by a 1×1 convolution kernel whose number of channels is set to be half the number of input channels.
θ_{p}	Feature θ_{p} obtained by θ through the SPP model.
X	Size of the convolution kernel.
N	Size of the feature map before SPP sampling.
P	Sampling model of SPP.
"$+x_i$"	Perform a residual operation on the input x_i.
W_z	It is represented by a 1×1 convolution kernel with the same number of channels as the number of input channels.
Z_i	Output features of asymmetric pyramid non-local block (APNB).
U	Multiscale fusion feature through the Inception structure.
F_{sq}	Squeeze operation.
F_{ex}	Excitation operation.
$\tilde{U}$	Output features of inception and squeeze-and-excitation block (ISEB).
F_{scale}	Scale operation on U feature weight.
C_{in}, C_{out}	Number of output channels of the previous layer and this layer.
M	Size of the feature map after SPP sampling.
r	Dimensionality reduction ratio in W_1, which is set to be 16.
δ	Relu function.
W_1, W_2	W_1 and W_2 are selected as fully connected layers for dimensionality reduction and dimensionality increasing, which can reduce the amount of computation.
S	Weight 0∼1.

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I. INTRODUCTION

SAR is a high-resolution coherent imaging radar [1]. Compared with infrared, visible light, and other technologies, SAR can work around the clock in all weather conditions [2]. It can also monitor and detect targets covered by vegetation

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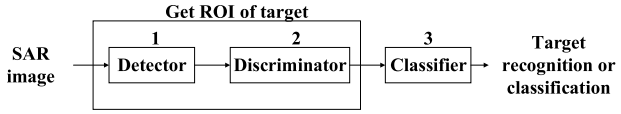


Fig. 1. Traditional SAR ATR system.

and sand with the help of the penetrating characteristics of the microwave. At the same time, SAR uses synthetic aperture and pulse compression technology to solve the contradiction between resolution and detection range. In addition, SAR has also multiple working modes, such as multipolarization and multiband. It can obtain various electromagnetic characteristics of the ground scene in all directions [3]. Based on these advantages, SAR image target recognition technology is widely used in various fields, including agriculture and forestry management, military target monitoring, and disaster prevention. Automatic target recognition (ATR) can distinguish targets or other objects using data from sensors, and it has a wide range of applications [4].

Detection, discrimination, and recognition are the three major steps in the traditional SAR ATR [5]. The traditional SAR ATR is shown in Fig. 1. The first step is the target detection. The purpose is to separate the area that may contain a specific target from the SAR image that includes a lot of background clutter. Constant false-alarm rate (CFAR) detector [6] is a classic method to detect, which determines the detection threshold according to the given false alarm rate. However, for SAR images with complex backgrounds, the detection probability of CFAR is greatly decreased. To remove outliers from SAR images with complex backgrounds while maximizing the preservation of true targets, a new two-parameter CFAR detector based on adaptively truncated clutter statistics (TS-LNCFAR) was proposed in [7]. To improve the detection performance of multitarget SAR images, an outliers-robust constant false-alarm rate (OR-CFAR) detector of Gaussian clutter based on the truncated-maximum-likelihood estimator (TMLE) was proposed in [8]. In [9], the robust CFAR detector based on bilateral-trimmed-statistics (BTS-RCFAR) improved the detection performance of multitarget SAR images in complex backgrounds. To reduce the interference of high-intensity noise and improve the detection performance, an automatic identification system (AIS) data-aided Rayleigh constant false alarm rate (AIS-RCFAR) was proposed in [10]. The second step is target discrimination. The detection results are further screened to eliminate false alarms caused by background clutter. Then, an image containing the target can be obtained. The most commonly used discrimination method is feature extraction. The performance of discriminator is directly affected by the feature detachability. Therefore, the research of this method mainly revolves around the design and extraction methods of identifying features [11]. The last step is target recognition. The small area images extracted by the detection and identification are further processed in this stage. The false targets are eliminated, and the target of interest category information is obtained [12]. Image preprocessing, feature extraction, and classification are the major processes in traditional target recognition systems. Image preprocessing enhances the target area and weakens the background area by removing coherent speckle noise, image segmentation, filtering, and other operations [13]. The common features for feature extraction include gray-level statistical features, geometric structure features, texture features, and electromagnetic scattering features. The quality of the extracted features affects the performance of SAR image recognition. Amrani et al. [14] proposed a feature extraction method based on a visual saliency model to highlight the ability of the most salient regions and improve the classification performance. The bag-of-visual-words (BoVW) algorithm accurately represents the target features and then combines the relevant features to obtain the discriminative features, which realizes high-precision, low-complexity SAR image classification [15], [16]. This article used Haar-like features to greatly improve the detection rate of SAR ship targets [17]. In [18], Bayesian-morphological saliency was used to detect high-resolution SAR images, which can not only highlight the target of interest but also guarantee the integrity of the target structure. Based on the significance and Gist (SG) features of optical satellite images, improved SG (MSG) features are proposed for SAR target identification [19]. The classifier maps feature to the target's category attributes. Common methods include support vector machines, logistic regression, and sparse representation. Traditional SAR image target recognition methods depend on the distinguishability of artificially designed features. For complex SAR images, traditional target recognition methods of artificially designed features are difficult to obtain better performance [20]. In addition, traditional methods lose a lot of image information in image preprocessing and feature extraction. The whole process of the traditional SAR ATR is complicated and cumbersome and has excellent limitations.

With the development of SAR technology, the imaged target contains more information. Therefore, further in-depth and detailed SAR image target detection and recognition studies are required. Recently, deep learning (DL) has achieved fruitful results in many areas, including face recognition, vehicle detection, and natural language processing [21]. DL not only extracts the important feature of different targets automatically but also reduces the time to select features and classifiers. The method based on DL can integrate the two steps of detection and identification. The target features are extracted through a deep neural network (DNN) with powerful feature expression capabilities, significantly reducing the workload. At the same time, it can avoid the limitations of artificially designing target features and significantly improve the detection and recognition capabilities of SAR image targets [11], [22]. In SAR image target detection, researchers have used transfer learning methods to make lots of attempts in SAR images. Then, they combine the characteristics of SAR image and DL technology to optimize and improve the method to adapt to the corresponding detection task [23]. At present, the mainstream feature extraction method based on DL is a convolutional neural network (CNN) [24]. CNN mainly performs feature transformation layer by layer on the original data. It can express the original data more abstractly at a higher level, making classification or prediction easier. For example, in 2012, Krizhevsky et al. [25] won the ILSVRC 2012 image classification competition with the AlexNet model,

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achieving a Top5 error rate of 15.3%. In the competition after ILSVRC, excellent models, such as the visual geometry group (VGG) [26], GoogLeNet [27], and residual network (ResNet) [28], have gradually emerged. CNN has been widely used for its powerful feature extraction capabilities. Amrani and Jiang [29] proposed a combined method of extracting deep discriminative features using VGG and K-nearest neighbor (KNN) classification to solve the SAR target classification problem. In [22], selective search was used for the first time to generate candidate regions. Meanwhile, CNN is used to classify the proposal regions. In [30], the SPP-Net model based on region-CNN (R-CNN) was proposed. This method effectively solves the limitation of CNN on the input scale and increases the running speed by about 100 times at the same time. To maintain the integrity of the detected SAR image target features, a multikernel size feature fusion convolutional neural network (MKSFF-CNN) was proposed in [31]. A new CNN model that maintains low-level texture and edge features and extracts high-level depth features was proposed in [32]. Based on the Faster-region CNN (RCNN) framework, Jiao et al. [33] proposed a densely connected multiscale neural network to improve SAR ship detection performance in complex offshore scenes. Tang et al. [34] combined DNN and extreme learning machine (ELM) to solve the problem of a long time and low detection performance for optical image detection. For the problem of SAR multisize target detection on the sea surface, the coupled CNN target detection network makes up for the shortcomings of Faster R-CNN in extracting target proposal regions [35]. In [19], a method based on visual attention was proposed, which can validly detect targets in high-resolution complex SAR images. In in-ground vehicle target detection, the progress is relatively slow due to the limitations of existing data. Some scholars use the method of embedding slice images of moving and stationary target acquisition and recognition (MSTAR) and min-SAR datasets into large scene images for research [36]. Subsequently, methods such as Mask R-CNN [37], YOLO [38], and single shot multibox detector (SSD) [39] appeared, and the recognition accuracy was further improved. Amrani et al. [40] proposed a robust feature extraction method for target classification in SAR images based on YOLOv4. The method adaptively fuses effective features from different CNN layers. To solve the problem of insufficient training samples for automatic classification of SAR targets based on DL, Yang et al. [41] proposed a mixed loss graph attention network (MGA-Net). This method has both robustness and higher accuracy. To simplify the network model and reduce the parameters, detection with transformers (DETR) eliminates the anchor generation frame and nonmaximum suppression (NMS) procedure [42]. Fully convolutional one-stage object detector (FCOS) eliminates predefined sets of anchor boxes and avoids the complex calculations associated with anchor boxes [43].

There are some problems with these methods. First, most DL network models have too many network parameters and calculations. As a result, the time cost of training the model is very high, and the real-time performance during inference is poor. Therefore, choosing a network architecture and setting key parameters are critical problems for improving network performance. Second, feature extractions based on convolution operations are all local operations in both space and time. The convolution kernel extracts the features in the local area with a specific step size. Increasing the number of convolutional layers and loops is the only method to obtain global information. This method, on the other hand, will make model training more difficult, network optimization more challenging, and even lead to over-fitting. Finally, SAR image target recognition methods primarily use the spatial information included in the feature map and neglect channel dependence.

In view of the shortcomings of the existing SAR image target recognition methods based on CNN, we propose a nonlocal channel attention network (NLCANet). First, we select GoogLeNet as the basic framework of the proposed method, which not only increases the width and depth of the network and improves the parameter utilization but also alleviates the problem of overfitting. Second, we propose to perform a traditional convolution on the input SAR image and input it to APNB for non-local (NL) operations. The purpose is to make the extracted features no longer limited to adjacent points and to extract more feature information. Finally, we propose to input the feature information extracted by APNB into the ISEB. The purpose is to obtain the channel dependencies established based on the fusion of different scale features and, simultaneously, alleviate the overfitting problem caused by APNB. The experiments verify that the detection and recognition of small targets in SAR image are improved by the NLCANet in complex scenes.

The remainder of this article is organized as follows. GoogLeNet is introduced in Section II. The proposed NLCANet framework is introduced in Section III. In Section IV, we perform experiments based on the MSTAR dataset and the SAR ship detection dataset (SSDD), and the experimental results demonstrate the effectiveness of the NLCANet for target recognition in SAR images. Finally, conclusions are drawn in Section V.

II. Related Work

A. GoogLeNet

With the increase in the convolutional layer number in traditional DL frameworks, there will be problems such as increased computational complexity, overfitting, and low accuracy. To solve this problem, the fully connected layer of the original framework was replaced with a sparse fully connected layer as proposed in [44]. Then, to reduce the enormous convolution operation, a 1×1 convolution layer is introduced. Finally, the Inception structure is designed by referring to the design idea of the "mlpconv" module in network in network (NIN) [45]. The advantage of Inception is that it can expand the size of the layer domain without causing computational difficulties. The Inception structure is shown in Fig. 2.

Fig. 3 shows the GoogLeNet framework composed of feature extractors and classifiers. The convolution layer, pool layer, and nine Inception V1 make up the feature extractor. The classifier is made up of fully connected layer and dropout. Two auxiliary classifiers are added to the GoogLeNet after Inception (4a) and Inception (4d) for forward propagation

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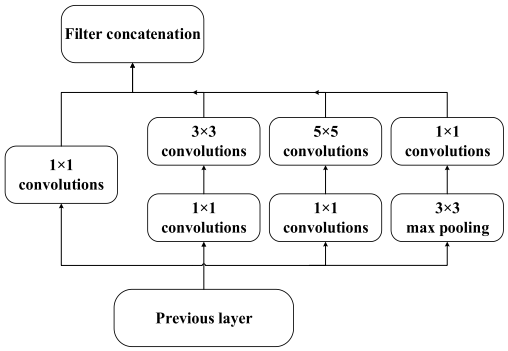


Fig. 2. Structure of Inception.

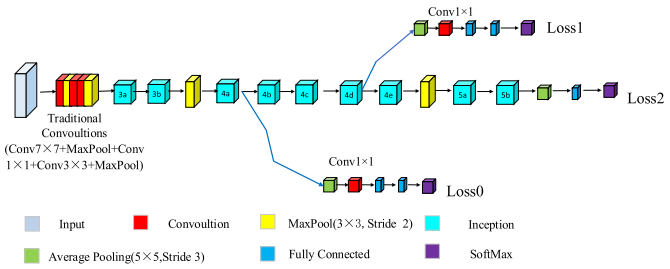


Fig. 3. Detailed block diagram of GoogLeNet.

gradients, respectively. Then, the auxiliary classification loss is added to the final classification result with a small weight (0.3). Softmax is used for GoogLeNet classification loss function. Formally, the input features x_i have corresponding labels y_i, and the softmax loss can be written as Loss $= (1/N')\sum_{i=1}^{K} - \log(e^{f_{yi}}/(\sum_j e^{f_j}))$. K represents the number of classes, N' represents the number of training data, and f represents the activation function of the fully connected layer W. f_j is the jth element of the class score f, $f_{y_i} = W_{y_i}^{\mathrm{T}} x_i$. The final fusion classification result is Loss $=$ Loss2 $+ 0.3$Loss1 $+$ 0.3Loss0. Loss0 and Loss1 are the classification results of the auxiliary classifier, and Loss2 is the final classification result. It not only improves the discriminative power of lower stage features, but it also solves the vanishing gradient problem and provides regularization. The performance of GoogLeNet is prominent in DL. It increases the depth and width of the network while reducing the convolution operation's computation and keeping the network computation invariant [46].

B. Attention Mechanism

The attention mechanism in DL is similar to the selective visual attention mechanism in humans. By scanning the global image, areas of information about the target are focused on and the attention of other information is reduced [47]. In recent years, attention mechanism has been widely used in target detection tasks with good results. The attention mechanism solves the information overload problem and improves the efficiency and accuracy of task processing. Squeeze-and-excitation network (SENet) [48] is the classical channel attention mechanism, which enhances the representation of the network by learning the dependencies between channels. Convolutional block attention module (CBAM) [49] combines spatial and channel attention mechanisms to effectively improve CNN representation by focusing on important feature information and suppressing irrelevant feature information. Self-attention mechanism is a branch of attention mechanism. Wang et al. [50] first proposed the non-local module in 2018. The interaction between any two locations can be computed by NB to capture the remote dependencies directly, rather than being limited to adjacent points. The purpose is to extract rich global information that allows the network to identify targets more accurately and efficiently.

III. Non-Local Channel Attention Network

A. Proposed Method

The proposed method adds APNB after the second max-pooling layer and SEB after each Inception of the GoogLeNet framework. The structure of the NLCANet is shown in Fig. 4. In Fig. 4, the NLCANet classification network frame is in the red dashed box, and the NLCANet detection and recognition network frame is in the green dashed box.

The proposed NLCANet based on GoogLeNet mainly includes two modules. First, the SAR image after traditional convoluted is input to APNB for non-local operation. The interaction between any two locations can be calculated by APNB to directly capture the remote dependence, rather than being limited to adjacent points. The purpose is to extract abundant feature information. By enhancing segmentation performance, APNB increases NB learning capacity and simultaneously lowers its computational and memory overhead. Second, the feature information extracted by APNB is input into ISEB. The informative features previously obtained by Inception are recalibrated by SEB through a squeeze-and-excitation operation in ISEB. The purpose is to obtain the channel dependencies established based on the fusion of different scale features. Meanwhile, the overfitting problem caused by APNB is alleviated. In the SAR image classification experiment, the outputs of Inception (4a) and Inception (4d) in NLCANet are used as auxiliary classification and fused with the final classification loss. In the SAR image target detection and recognition experiment, NLCANet draws on the idea of feature pyramid network (FPN) to extract Inception (4a), Inception (4d), and Inception (5b) three-layer feature maps. The small feature map is then upsampled to the same size as the large feature map and spliced with it for further prediction. After that, the nine scales of anchor boxes are uniformly distributed to the feature maps of three scales using the concept of dimensional clustering. Then, redundant prediction boxes are removed by NMS. Finally, the detection and recognition results are obtained. The loss function of detection and recognition consists of box loss, object loss, and class loss. In this article, it is a single-class target detection experiment for ships, so class loss is 0. BCEWithLogitsLoss is used for object loss, and generalized intersection over union (GIOU) is used for box loss.

The non-local channel attention architecture is composed of APNB and ISEB. In this article, APNB and ISEB are described in Sections III-B and III-C, respectively.

B. Asymmetric Pyramid Non-Local Block

The convolutional operation of the CNN process is a local neighborhood. If we want to get remote dependencies,

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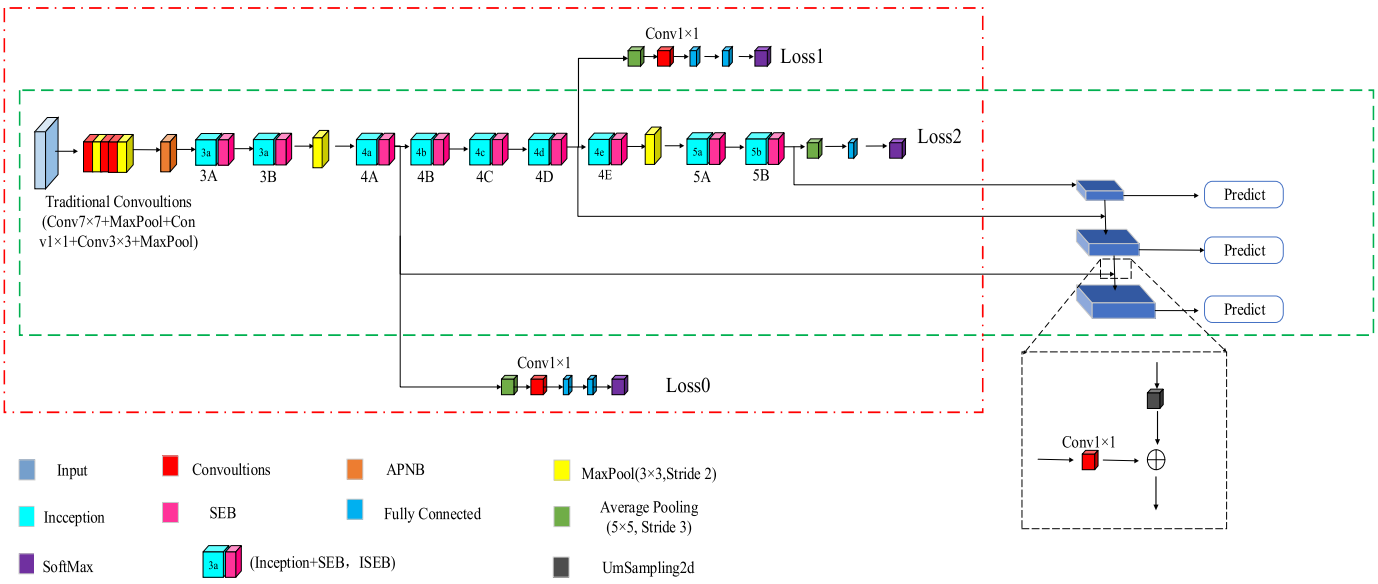


Fig. 4. Structure of the NLCANet.

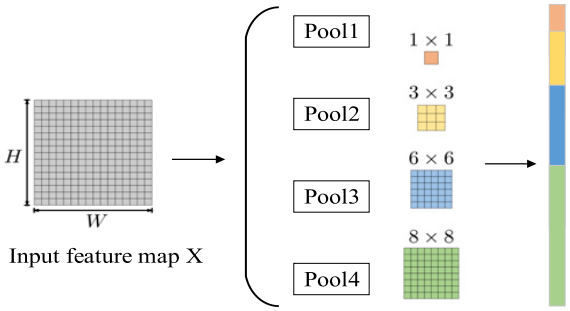


Fig. 5. Detailed block diagram of SPP.

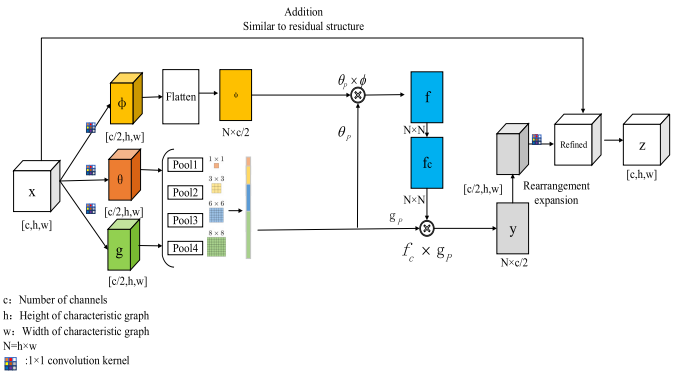


Fig. 6. Detailed block diagram of APNB.

it can only be done by repeating local operations and passing data gradually. This method will make model training more difficult, network optimization more challenging, and even cause over-fitting. An NB based on NL-mean [51] in the field of image denoising and the self-attention [52] method for processing time-series information is proposed in [50]. However, NB still consumes a lot of time and memory. SPP is used to reduce the computational cost of NB matrix complications [53]. The structure of SPP is shown in Fig. 5. This is equivalent to selecting several representative points from θ and g instead of calculating all representative points. The interaction between any two locations can be calculated by APNB to directly capture the remote dependence, rather than being limited to adjacent points. The purpose is to extract abundant feature information. The detailed block diagram of APNB is shown in Fig. 6.

Therefore, the general formula for non-local operations is written as

$$y_i = \frac{1}{C(x)} \sum_{\forall j} f\left(x_i, x_j\right) g_{\mathrm{p}}\left(x_j\right) \tag{1}$$

where x $(h \times w \times c)$ represents the feature map obtained by the traditional convolutions of the input SAR image. The width, height, and number of channels of the feature map are represented by w, h, and c, respectively. x_i denotes the input pixel, y_i denotes the response of output pixel, and x_j denotes the possible neighboring pixels of all pixels in the image. The response of the input signal x_j is computed by g_{p}

$$g_{\mathrm{p}}\left(x_j\right) = P\left(W_{\mathrm{g}} x_j\right) \tag{2}$$

where W_{g} is the same as the weight of automatic learning and P is the sampling model of SPP. $C(x)$ is the sum of all $f(x_i, x_j)$ that used to normalize the response. The function f is an embedded Gaussian function, which calculates the similarity between pixels x_i and x_j. The function f is written as

$$f\left(x_i, x_j\right) = e^{\theta_{\mathrm{p}}(x_i)^{\mathrm{T}} \varphi\left(x_j\right)} \tag{3}$$

where $\theta_{\mathrm{p}}(x_i) = P(W_\theta x_i)$ and $\varphi(x_j) = W_\varphi x_j$. W_θ and W_φ are the weights automatically learned by the network. It can be seen that for a given x_i in (3), $(1/C(x)) \sum_{\forall j} f(x_i, x_j)$ is the calculation of all softmax, which is written as

$$y_i = \mathrm{softmax}\left(\left(P_\theta(W_\theta x_i)\right)^{\mathrm{T}} W_\varphi x_j\right) g_{\mathrm{p}}\left(x_j\right). \tag{4}$$

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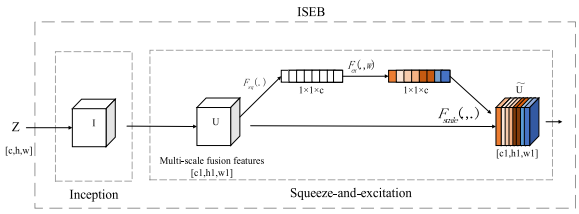


Fig. 7. Detailed block diagram of ISEB.

To convert the non-local operation to APNB, the equation is written as

$$Z_i = W_z y_i + x_i \tag{5}$$

where W_z is the learnable weight, "$+x_i$" represents the residual operation, and Z_i is the output of APNB. It uses the principle of the ResNet to reduce the amount of calculation. Therefore, W_g, W_φ, and W_θ are represented by a 1×1 convolution kernel for actual operation, and their channel number is set to half of the channel number [28]. This parameter setting reduces the computational cost by 1/2. Finally, W_z is designed as a 1×1 convolution kernel whose purpose is to ensure that APNB can make output and input the same size ($h \times w \times c$).

C. Inception and Squeeze-and-Excitation Block

To prevent overfitting caused by APNB extracting too much noise information, we input the feature information obtained by APNB into ISEB. ISEB is composed of SEB and Inception in GoogLeNet. The detailed block diagram of ISEB is shown in Fig. 7. The Inception structure not only reduces the amount of computation, improves the recognition accuracy, and alleviates overfitting, but also obtains multiscale fusion features. SEB aims to obtain the relationship between channels in Inception and then adjusts the weight of the channels [48]. SEB mainly includes squeeze and excitation operations. The multiscale fusion feature $U \in R^{c\times h\times w}$ is obtained from the output $Z \in R^{c\times h\times w}$ of APNB through the Inception structure. Then, $F_{\text{sq}}(.)$ performs a squeeze operation on U. The dimension of the U feature vector is not changed, but the vector value has become a new value. Then, the weight is generated for each feature channel through the excitation operation $F_{\text{ex}}(., W)$. Finally, U is weighted by $F_{\text{scale}}(.,.)$ to get $\tilde{U}$.

The squeeze operation obtains the dependencies between channels through a global average pool. It is to convert the 2-D $h_1 \times w_1$ feature into a real number with a global receptive field [54]. The squeeze operation is expressed as

$$a_c = F_{\text{sq}}(U) = \frac{1}{h_1 \times w_1} \sum_{i=1}^{h_1} \sum_{j=1}^{w_1} U(i, j) \tag{6}$$

where U denotes the multiscale fusion feature, which is $c_1 \times h_1 \times w_1$. The width, height, and the number of channels of feature map are represented by w_1, h_1, and c_1, respectively.

The excitation operation is comprised of two major steps. The first part is a gate mechanism. First, the first fully connected layer reduces the dimensionality of the c real numbers by r times. Then, use the Relu function to perform nonlinearization. It uses the Relu function as the activation function to make the structure more capable of learning the nonlinear interrelationships between channels. Finally, the second fully connected layer multiplies the dimension by r. The second part is to first use the sigmoid function to estimate the weight of the channel. Then, adjust each channel according to the weight estimation through the scale operation. Finally, we get into the next layer of the network [55]. The formula for using the sigmoid function to obtain the weight 0∼1 is written as

$$S = F_{\text{ex}}(a_c, W) = \sigma(g(a_c, W)) = \sigma(W_2 \delta(W_1 a_c)) \tag{7}$$

where δ denotes the Relu function, $W_1 \in R^{(c/r)\times c}$, and $W_2 \in R^{c\times(c/r)}$. We optimize the gating mechanism by W_1, Relu, and W_2. Specifically, W_1 and W_2 are selected as fully connected layers for dimensionality reduction and dimensionality increasing, which can reduce the amount of computation. It uses the Relu function as the activation function to make the structure more capable of learning the nonlinear interrelationships between channels. r is the dimensionality reduction ratio in W_1, which is set to be 16.

The feature U is weighted by the scale operation to get $\tilde{U}$. Its calculation formula is written as

$$\tilde{U} = F_{\text{scale}}(U, S) = SU \tag{8}$$

where $\tilde{U} = [\tilde{u}_1, \tilde{u}_2, \tilde{u}_3, \ldots, \tilde{u}_c]$. $U = R^{h\times w}$ is multiplied by S to get $F_{\text{scale}}(U, S)$.

D. Analysis of the Complexity of the Proposed Method

Increasing the depth and width of a CNN is the most obvious way to improve its performance. However, this method would add an excessive expense of CNN parameters and computational complexity, making it difficult to use. The 1×1 convolution is added to the 3×3 and 5×5 convolution branches in the Inception structure, which decreases the time complexity of Inception by three times and reduces the number of parameters from 1000 to around 300 K. Simultaneously, the fully connected layer in GoogLeNet is replaced by global average pooling, and the complexity is no longer affected by the size of the input image. Its time complexity is changed from $O(1^2 \cdot X^2 \cdot C_{\text{in}} \cdot C_{\text{out}})$ to $O(C_{\text{in}} \cdot C_{\text{out}})$, where X denotes the size of the convolution kernel, and C_{in} and C_{out} denote the number of output channels of the previous and current layer, respectively. The primary computational expense in NB is the matrix multiplication of $\theta \times \varphi$ and $f_c \times g$. SPP is used for APNB to sample feature map size from N to M ($M \ll N$). $N = h \times w$ and $M = h' \times w'$ denote the size of the feature map before and after SPP sampling, respectively. It not only keeps the output size of the two matrices commensurate but also changes the computational complexity from $O((c/2)N^2)$ down to $O((c/2)S^2)$. W_1 and W_2 are used for the excitation operation in SEB to reduce the number of channels and thus reduce the amount of computation. Thought NLCANet deepens the depth and width of the network, it improves the recognition accuracy and avoids the problem of too many parameters to reduce the computational complexity.

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TABLE I
EXPERIMENTAL SETUPS

Name	Configuration
CPU	I9-10920X
RAM	64GB
GPU	RTX3080 TI
Operating system	Windows10
Deep learning platform	Pytorch

IV. EXPERIMENT AND ANALYSIS

A. Evaluation Metrics and Experimental Setup

In this article, the evaluation metrics of the SAR image target recognition result include the following contents.

1) In the identical setup environment, training set, and testing set, Accuracy, Precision, Recall, F1, and mean average precision (MAP) for SAR image target recognition are expressed as

$$\text{Accuracy} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{9}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{10}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{11}$$

$$\text{F1} = \frac{2\times\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}} \tag{12}$$

$$\text{MAP} = \text{AP} = \int_0^1 P(R)dR. \tag{13}$$

F1 is the harmonic average of Precision and Recall rate, which can measure the comprehensive performance of the detector. Its value is between 0 and 1, and the closer to 1, the better the detector performance, where MAP, P is the precision rate, R is the recall rate, and $P(R)$ is the P–R curve of the precision–recall rate. The average accuracy AP is the area under the P–R curve, which can be used to evaluate the overall performance of the detector. Because of the single-target ship detection in this article, the average accuracy AP is equal to the class average accuracy MAP. The higher the MAP value is, the better the performance of the detector.

2) The time it takes to run for different methods.
3) In the same setup environment, the training loss curve and testing accuracy curve of different methods.
4) SAR image prediction results under complex background.

The experimental setups are shown in Table I.

B. Result of Land Targets Recognition

As the basic framework of the proposed method, GoogLeNet has advantages in SAR image classification and recognition in complex scenes. Therefore, the MSTAR dataset of a single target scene from the Sandia National Laboratory, USA, [56] is used to compare the proposed method with other methods. Examples of SAR images and optical images of ten target classes of targets in the MSTAR dataset are shown in Fig. 8. The number of training and testing samples of the MSTAR dataset in this article is shown in Table II. The MSTAR dataset has a total of 6874 images. The images under 17° depression angle are used for training, and the images under 15° depression angle are used for testing. For the training process on the MATAR dataset, we set the learning rate as 0.0003, epoch as 100, and batch_size as 16 to obtain the optimal performance.

1) Ablation Study Based on GoogLeNet: The ablation study based on GoogLeNet is shown in Table III. After the second, third, and fourth max-pooling layers of the GoogLeNet framework, APNB is added and denoted as GoogLeNet-preAPNB, GoogLeNet-midAPNB, and GoogLeNet-postAPNB methods,

TABLE II
MSTAR DATASET: NUMBER OF TRAINING AND TESTING SAMPLES

Target	Train		Test	
	Depression	No.Image	Depression	No.Image
SN-C21	17°	233	15°	196
BMP-2 SN-C9563	17°	233	15°	195
SN-C9566	17°	232	15°	196
BTR-70	17°	233	15°	196
SN-123	17°	232	15°	196
T-72 SN-812	17°	231	15°	195
SN-S7	17°	228	15°	191
BTR-60	17°	256	15°	196
2S1	17°	299	15°	274
BRDM-2	17°	299	15°	274
D7	17°	299	15°	274
T-62	17°	299	15°	274
ZIL-131	17°	299	15°	274
ZSU-234	17°	299	15°	274
Total		3671		3202



Fig. 8. Types of the target in MSTAR: (left) SAR image and (right) optical images.

TABLE III
ABLATION STUDY BASED ON GOOGLENET

Method	Accuracy (%)
GoogLeNet	99.16
GoogLeNet-preAPNB	99.53
GoogLeNet-midAPNB	99.38
GoogLeNet-postAPNB	99.22
GoogLeNet-ISEB	99.34
Proposed method	**99.72**

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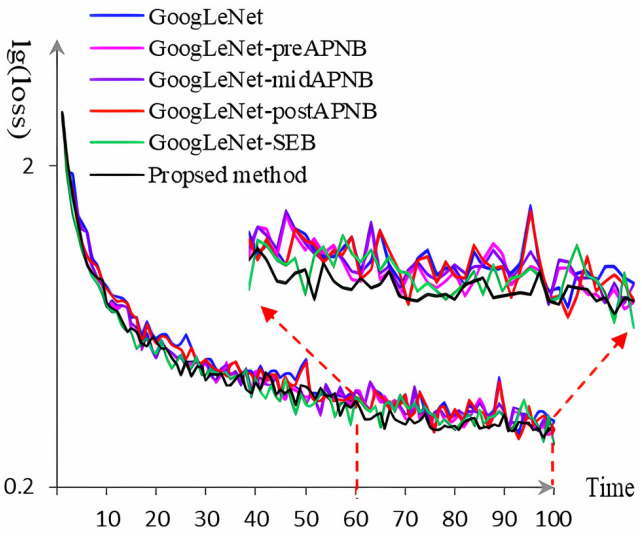


Fig. 9. Training loss curve of the proposed method and other GoogLeNet methods.

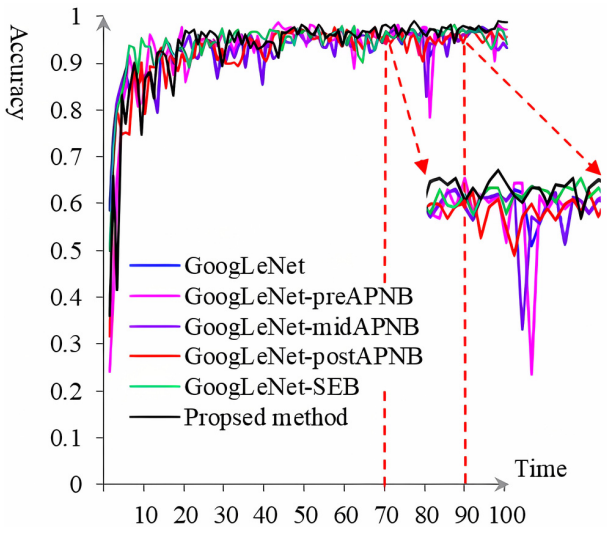


Fig. 10. Testing accuracy curves of the proposed method and other GoogLeNet methods.

respectively. After the Inception of the GoogLeNet framework, the SEB is added to form the ISEB and denoted as the GoogLeNet-ISEB method. It can be seen from Table III that adding APNB or SEB to the GoogLeNet framework improves the recognition accuracy. In the experiment of adding APNB at different positions of the GoogLeNet framework, GoogLeNet-preAPNB improved the most, reaching 99.53%. The main reason is that the further ahead the APNB is, the more feature information from the SAR image is obtained. Therefore, SEB is added to GoogLeNet-preAPNB to further improve the recognition accuracy to 99.72%. The ten military targets of Precision, Recall, and F1 based on GoogLeNet methods in the MSTAR dataset are shown in Tables IV and V, respectively. The maximum value for each type of target is shown in black bold in Tables IV and V. From Tables IV and V, Precision, Recall, and F1 of ten categories of military targets are significantly improved by the proposed method.

The training loss curves of the proposed method and other GoogLeNet methods are shown in Fig. 9. From Fig. 9, the training loss curves of the six methods all show a downward trend and eventually stabilize. During 0–60 times, the training loss curves of the six methods are similar. During 60–100 times, the training loss curve of the proposed method is significantly lower than the other five methods. Therefore, adding APNB and SEB to the GoogLeNet framework strengthens the feature extraction of SAR image targets and reduces information loss during the training process.

The testing accuracy curves of the proposed method and other GoogLeNet methods are shown in Fig. 10. From Fig. 10, the testing accuracy curves of the six methods all show an upward trend and eventually tend to stabilize. Nevertheless, the testing accuracy curve of the proposed method is better than that of other methods. During 70–90 times, the fluctuation of the testing accuracy curve of GoogLeNet-preAPNB and Goog-LeNet-midAPNB is larger than that of GoogLeNet, GoogLeNet-postAPNB, and GoogLeNet-SEB, and the proposed method. The reason is that the MSTAR dataset is too small, and the SAR image contains too much noise. APNB is added in the front part of the GoogLeNet framework to learn noise rules while obtaining more feature information. It causes overfitting of the accuracy of the test results. From Table III and Fig. 10, SEB is added to GoogLeNet-preAPNB to enhance the recognition accuracy and resolve the problem of over-fitting.

2) Experimental Analysis of the Superiority of APNB and ISEB: The experimental analysis of the superiority of the APNB and ISEB with the existing atrous spatial pyramid pooling (ASPP) [57] and dense block (DB) [58] is shown in Table VI. From Table VI, the classification accuracy of GoogLeNet-APNB is 0.15% greater than that of GoogLeNet-ASPP. The interaction between any two locations can be calculated by APNB to capture the correlation between long-range features. The atrous convolution is used in ASPP, so there will be a loss of local information between features and a lack of correlation between the information obtained at a distance. The classification accuracy of GoogLeNet-ISEB is 5.4% higher than that of GoogLeNet-DB. There are two auxiliary classifiers in the network framework of the proposed method, which not only strengthens the use of intermediate layer features but also avoids gradient disappearance. ISEB obtains channel dependencies based on feature fusion at different scales. The output of each layer in the DB is directly connected to all subsequent layers, which not only strengthens the reuse of target features but also strengthens the reuse of noise information, resulting in poor classification results. Compared with GoogLeNet-ASPP, the classification accuracy of GoogLeNet-ASPP-ISEB is reduced by 0.1%. Because the images of the MSTAR dataset are small and monotonous, the local feature information will be lost after processing by ASPP, and then extracting features by ISEB will reduce the accuracy. From the experimental results, it can be seen that APNB and ISEB of the proposed method have more advantages than ASPP and DB.

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TABLE IV
PRECISION AND RECALL OF TEN TYPES OF MILITARY TARGETS IN MSTAR BASED ON GOOGLENET METHODS

Type	Precision（%）						Recall（%）					
	GoogLeNet	GoogLeNet-preAPNB	GoogLeNet-midAPNB	GoogLeNet-postAPNB	GoogLeNet-SEB	Propsed method	GoogLeNet	GoogLeNet-preAPNB	GoogLeNet-midAPNB	GoogLeNet-postAPNB	GoogLeNet-SEB	Propsed method
2S1	98.5	98.2	**100**	99.6	99.6	99.6	98.2	99.3	**100**	97.8	**100**	**100**
BMP2	99.0	99.7	98.5	99.7	99.5	**99.8**	98.8	99.3	99.5	**99.5**	99.1	**99.5**
BRDM_2	**100**	**100**	99.6	**100**	**100**	**100**	98.9	99.6	**100**	99.3	**100**	99.6
BTR70	99.0	**100**	**100**	95.6	98.5	98.5	**100**	98.0	98.0	**100**	99.5	**100**
BTR_60	97.5	99.5	98.5	**100**	97.5	98.6	**100**	97.9	99.0	94.4	98.5	**100**
D7	**100**	**100**	99.6	99.3	99.6	**100**	**100**	**100**	99.6	99.3	**100**	**100**
T62	98.6	**100**	**100**	98.9	**100**	**100**	99.6	**100**	99.6	99.6	99.3	**100**
T72	**100**	99.6	99.3	99.1	**100**	**100**	98.5	99.6	98.8	**99.7**	**99.7**	**99.7**
ZIL131	98.6	99.3	**99.6**	98.2	99.5	**99.6**	**100**	**100**	99.6	**100**	**100**	**100**
ZSU_23_4	**100**	99.6	99.6	99.6	**100**	**100**	**100**	**100**	99.6	**100**	**100**	98.9

TABLE V
F1 OF TEN TYPES OF MILITARY TARGETS IN MSTAR BASED ON GOOGLENET METHODS

Type	F1(%)					
	GoogLeNet	GoogLeNet-preAPNB	GoogLeNet-midAPNB	GoogLeNet-postAPNB	GoogLeNet-SEB	Propsed method
2S1	98.3	98.7	**99.8**	98.9	**99.8**	**99.8**
BMP2	98.9	**99.5**	99.0	99.4	99.3	**99.5**
BRDM_2	99.4	99.8	99.8	99.6	**100**	99.8
BTR70	99.5	99.0	99.0	97.8	99.0	**100**
BTR_60	98.7	98.7	98.7	97.1	98.0	**99.2**
D7	**100**	**100**	99.6	99.3	99.8	99.3
T62	99.1	**100**	99.8	99.2	99.6	**100**
T72	99.2	99.6	99.0	99.4	**99.8**	**99.8**
ZIL131	99.3	99.6	99.6	99.1	99.4	**99.8**
ZSU_23_4	99.4	99.6	99.6	99.8	**100**	99.4

3) Experimental Analysis of the Proposed Method and the State-of-the-Art Machine Learning Methods: Table VII shows the classification result of the proposed method and different state-of-the-art machine learning methods, including directed acyclic graph support vector metric learning (DAG-SVML) [14], support vector metrics based on BoVW using histogram of oriented gradients discriminant correlation analysis (HBD-SVM) [15], and Gabor BoVW support vector metric (GB-SVM) [16]. From Table VII, the proposed method has higher recognition accuracy than other state-of-the-art machine learning methods. Compared with the state-of-the-art machine learning methods, the classification time of the proposed method takes a lot of time. The reason is that the network structure of the proposed method is complex, which requires traditional convolution, APNB and ISEB for feature extraction, and Softmax layer for classification. The confusion matrix of the classification results of the proposed method and the state-of-the-art machine learning methods is shown in Fig. 11. From Fig. 11, compared with the state-of-the-art machine learning methods, the number of correct classifications for BRDM_2,

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TABLE VI
APNB AND ISEB SUPERIORITY COMPARISON EXPERIMENTS

Method	Accuracy (%)
GoogLeNet	99.16
GoogLeNet-APNB	99.53
GoogLeNet-ASPP	99.38
GoogLeNet-ISEB	99.34
GoogLeNet-DB	93.94
GoogLeNet-APNB-DB	96.43
GoogLeNet-ASPP-ISEB	99.28
GoogLeNet-ASPP-DB	95.13
Proposed method	**99.72**

TABLE VII
CLASSIFICATION RESULTS OF THE PROPOSED METHOD AND MACHINE LEARNING METHODS

Method	Accuracy (%)	Testing time(ms)
DAG-SVML	99.68	23
HBD-SVM	99.52	2090
GB-SVM	99.46	280
Proposed method	**99.72**	9595

TABLE VIII
CLASSIFICATION RESULTS OF THE PROPOSED METHOD AND CNN METHODS

Method	Accuracy (%)	Training time(s)	Testing time(ms)
AlexNet	96.78	2400	9296
Vgg16	91.66	4500	11320
GoogLeNet	99.16	2900	9350
ResNet50	99.09	3700	9760
MDCA-CNN	95.66	5079	9813
VGG-KNN	99.82	86400	271400
Swin-Transform	96.34	5880	11663
Proposed method	99.72	3300	9595

BTR70, ZIL131, and ZSU_23_4 is significantly improved by the proposed method.

4) Experimental Analysis of the Proposed Method and CNN Methods: Table VIII shows the classification results of the proposed method and different CNN methods such as AlexNet [25], Vgg16 [26], GoogLeNet [27], ResNet50 [28], maximum density clustering application on convolutional neural network (MDCA-CNN), a combined method for SAR target classification using KNN on deep discriminative features extracted by VGG (VGG-KNN) [29], and Swin-transform [59]. From Table VIII, the proposed method outperforms other CNN methods. Compared with the deep feature extraction method ResNet50, the recognition accuracy of the proposed method is improved by 0.63% while decreasing training and testing times by 100 s and 165 ms, respectively. Compared with GoogLeNet, the recognition accuracy of the proposed method is increased by 0.56% while only increasing training time and testing time by 300 s and 245 ms, respectively. The recognition accuracy of the proposed method is slightly as good as VGG-KNN, but the training time and testing time are only 1/26 and 1/28 of VGG-KNN, respectively. Because VGG is used to extract deep features during training and KNN compares the test sample with each training sample during testing, the training and testing efficiency of VGG-KNN is significantly decreased. Compared with Swin-transformer, the recognition accuracy of this method is improved by 3.38%, and the training time and test time are reduced by 2580 s and 2068 ms, respectively. Due to the small size of the MSTAR dataset, the classification performance of the Swin-transformer is poor. Moreover, the Swin-transformer requires too much computation and storage, resulting in long training and testing times. Compared with MDCA-CNN, the proposed method not only improves the recognition accuracy, but also reduces the training and testing time. The main reason is that the MDCA-CNN method has problems such as large amount of calculation, slow classification speed, and difficulty in optimization. At the same time, MDCA is not suitable for datasets with large density differences, which easily cause the loss of feature information. However, the proposed method selects GoogLeNet as the basic framework, combined with APNB and SEB, which has the advantages of increasing the width and depth of the network, maintaining more feature information, improving the recognition accuracy, improving the utilization of parameters, and saving training time.

The ten military targets of Precision, Recall, and F1 of the eight CNN methods in the MSTAR dataset are shown in Tables IX and X, respectively. The maximum value for each type of target is shown in black bold in Tables IX and X. From Tables VIII and IX, the classification of ten categories of military targets by the proposed method and VGG-KNN is significantly improved compared to the other CNN methods.

The above experimental results show that the proposed method has certain advantages over the state-of-the-art machine learning methods and DL methods for target recognition on the MSTAR dataset.

C. Result of Detection and Recognition of Ship Targets on the Sea Surface

To further verify the performance of the proposed method, SSD, YOLOv3, Faster-RCNN, YOLOv4, GoogLeNet, FCOS, and DETR are selected for comparative experiments. The complex scene SSDD [60] is used in the experimental data. The SSDD has 1160 images and 2456 ship targets, with an average of 2.12 ships in each image. During the training process, the training set, testing set, and validation set consist of 939 pictures, 116 pictures, and 105 pictures, respectively. In this article, the training epoch is set to 200. The first 100 times

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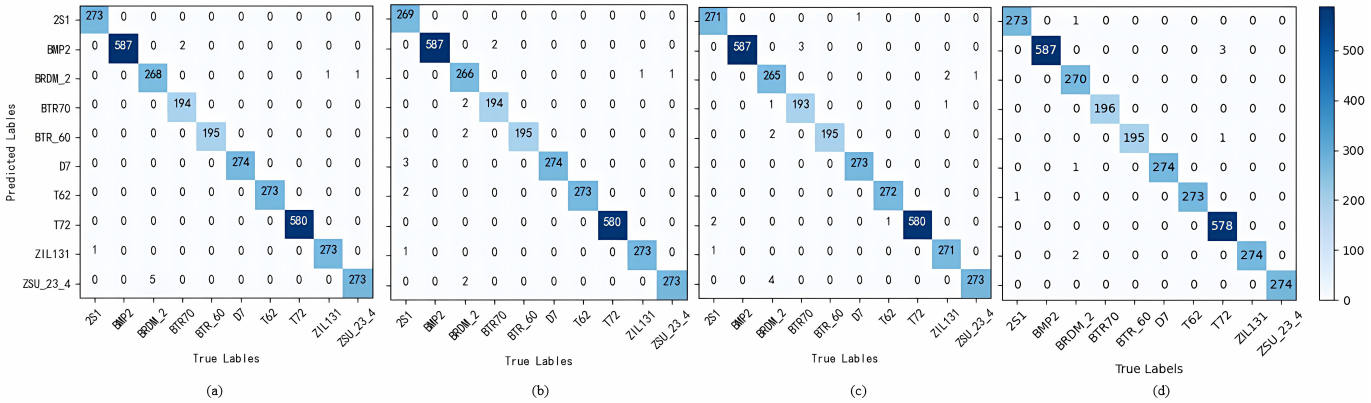


Fig. 11. Confusion matrix comparison between the proposed method and state-of-the-art machine learning methods: (a) DAG-SVM; (b) HBD-SVM; (c) GB-SVM; and (d) proposed method.

TABLE IX
PRECISION AND RECALL OF TEN TYPES OF MILITARY TARGETS IN MSTAR BASED ON CNN METHODS

Type	Precision (%)								Recall (%)							
	AlexNet	Vgg16	GoogLeNet	ResNet50	MDCA-CNN	VGG-KNN	Swin-transform	Propsed method	AlexNet	Vgg16	GoogLeNet	ResNet50	MDCA-CNN	VGG-KNN	Swin-transform	Propsed method
2S1	93.7	97.7	98.5	**99.6**	89.4	**99.6**	94.7	**99.6**	97.1	76.3	98.2	98.2	98.5	**100**	98.2	**100**
BMP2	94.5	96.3	99.0	99.8	93.3	98.4	96.6	**99.8**	99.0	94.4	98.8	98.0	96.8	**99.8**	96.4	99.3
BRDM_2	**100**	90.6	**100**	**100**	96.7	97.4	96.2	**100**	94.5	98.5	98.9	98.2	85.8	**99.6**	92.7	**99.6**
BTR70	96.4	95.6	99.0	**100**	97.7	**100**	93.9	98.5	94.4	88.8	**100**	99.5	86.2	97.7	93.9	**100**
BTR_60	95.3	89.4	97.5	97.0	97.4	**99.0**	98.3	98.6	92.8	82.1	**100**	98.5	94.4	99.4	90.3	**100**
D7	98.9	99.5	**100**	96.5	98.6	**100**	97.5	**100**	96.7	80.3	**100**	**100**	99.6	**100**	98.9	**100**
T62	96.0	97.8	98.6	97.8	97.7	**100**	92.7	**100**	97.1	94.5	99.6	**100**	91.9	**100**	98.2	**100**
T72	98.8	93.6	**100**	99.3	98.6	**100**	98.9	99.6	96.4	97.4	98.5	99.7	98.6	**100**	96.2	**99.7**
ZIL131	97.2	79.6	98.6	**99.6**	92.2	**99.6**	98.9	**99.6**	99.6	99.6	**100**	**100**	99.3	**100**	98.9	**100**
ZSU_23_4	98.1	94.2	**100**	99.6	97.1	97.6	92.7	**100**	96.7	**100**	**100**	99.4	97.8	98.2	97.1	98.9

TABLE X
F1 OF TEN TYPES OF MILITARY TARGETS IN MSTAR BASED ON CNN METHODS

Type	F1(%)							
	AlexNet	Vgg16	GoogLeNet	ResNet50	MDCA-CNN	VGG-KNN	Swin-transform	Propsed method
2S1	95.4	85.7	98.3	98.9	93.7	**99.8**	96.4	**99.8**
BMP2	94.5	95.3	98.9	98.9	95.0	99.1	96.5	**99.5**
BRDM_2	**100**	94.4	99.4	99.1	90.9	98.5	94.4	99.8
BTR70	96.4	92.1	99.5	99.7	91.6	98.8	93.9	**99.2**
BTR_60	95.3	85.6	98.7	97.7	95.9	**99.2**	94.1	**99.2**
D7	98.9	88.9	**100**	98.2	99.1	**100**	98.1	99.3
T62	96.0	91.0	99.1	98.9	94.7	**100**	95.4	**100**
T71	98.8	95.5	99.2	99.5	98.6	**100**	97.5	99.8
ZIL131	97.2	88.5	99.3	**99.8**	95.6	**99.8**	98.9	**99.8**
ZSU_23_4	98.1	97.0	**100**	99.4	97.4	97.9	94.8	99.4

were freezing training, and the backbone of the model was frozen. The feature extraction network does not change; it only fine-tunes the network. The following 100 times are thawing training, during which the feature extraction network and all the network parameters will change. Example images of SSDD are shown in Fig. 12.

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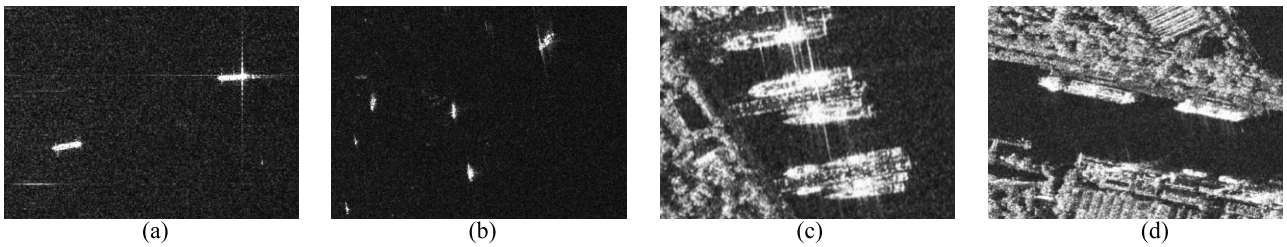


Fig. 12. Sample images of the SSDD. (a) and (b) SAR images of the ships under the background of the open sea. (c) and (d) SAR images of the ships in the inshore region.

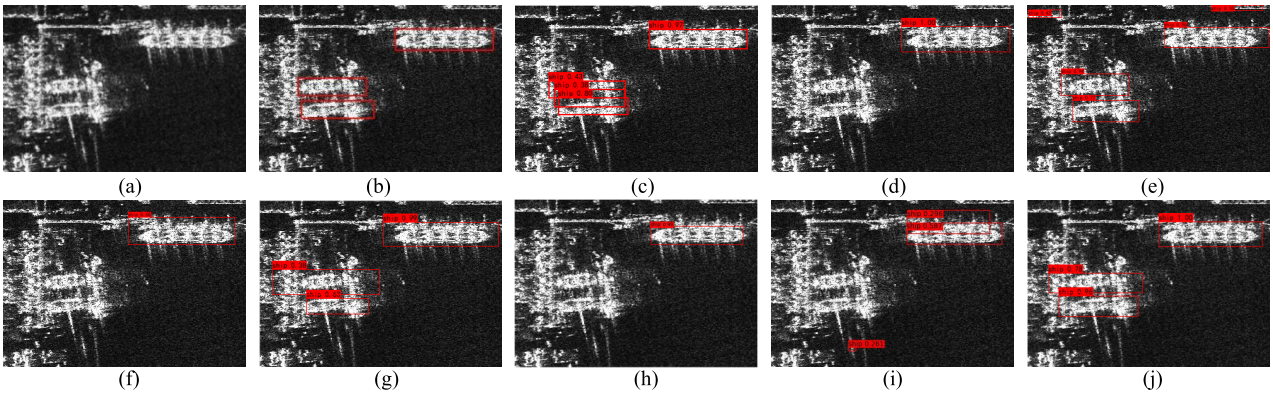


Fig. 13. Detection and recognition results of SAR images with three ships in the background of the inshore region. (a) Original image. (b) Accurate target position. (c) SSD. (d) YOLOv3. (e) Faster-RCNN. (f) YOLOv4. (g) GoogLeNet. (h) FCOS. (i) DETR. (j) Proposed method.

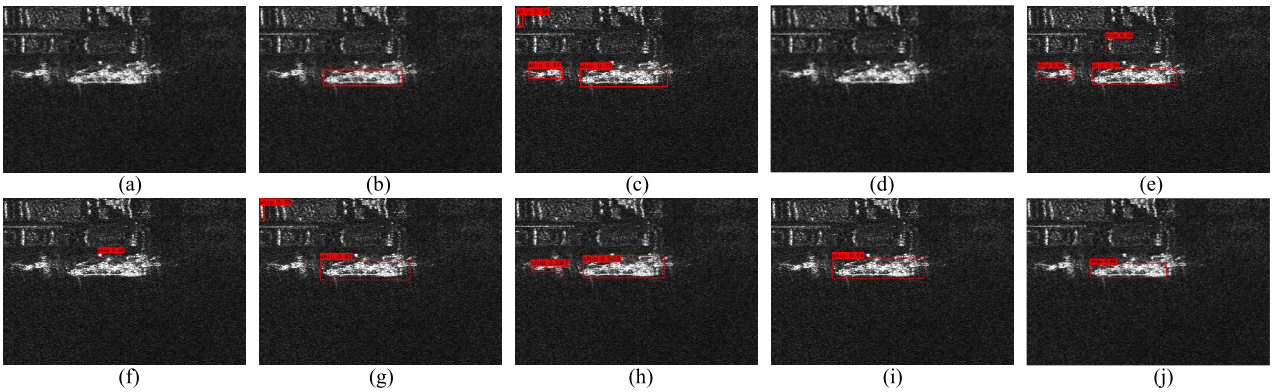


Fig. 14. Detection and recognition results of SAR images with one ship in the background of the inshore region. (a) Original image. (b) Accurate target position. (c) SSD. (d) YOLOv3. (e) Faster-RCNN. (f) YOLOv4. (g) GoogLeNet. (h) FCOS. (i) DETR. (j) Proposed method.

The target detection and recognition results with SSD, YOLOv3, Faster-RCNN, YOLOv4, GoogLeNet, FCOS, DETR, and the proposed method for the SSDD are shown in Figs. 13–16. In Figs. 13–16, the original image and the image including the accurate target position are shown in (a) and (b), respectively. The detection and identification results with SSD, YOLOv3, Faster-RCNN, YOLOv4, GoogLeNet, FCOS, DETR, and the proposed method are shown in Figs. 13(c)–(j)–16(c)–(j), respectively.

Figs. 13–15 are the SAR images in the background of the inshore and shore. The scattering intensity of the ship target is close to the coastal scene, so it is not easy to distinguish the ship target from the shore building. The missed detection and the false alarm are prone to occur. There are three real ship targets in Fig. 13. A total of four targets detected by SSD, which include three real ship targets and one nonship target. A target is correctly detected and recognized by YOLOv3, YOLOv4, and FCOS, but two ship targets are missed. A total of five targets detected by Faster-RCNN, which include three real ship targets and two nonship targets. The Faster-RCNN is a two-stage target test algorithm. Its first-level network is used to extract information in the candidate area, and the second-level network is used to classify and return to the candidate area. Therefore, it reduces the missing of ship targets in complex backgrounds. A total of three targets detected by DETR, which include a real ship targets and two nonship targets. Three targets correctly detected and recognized by GoogLeNet and the proposed method. Moreover, the proposed method outperforms GoogLeNet in terms of the recognition rate for the three ship targets. There is a real ship target in

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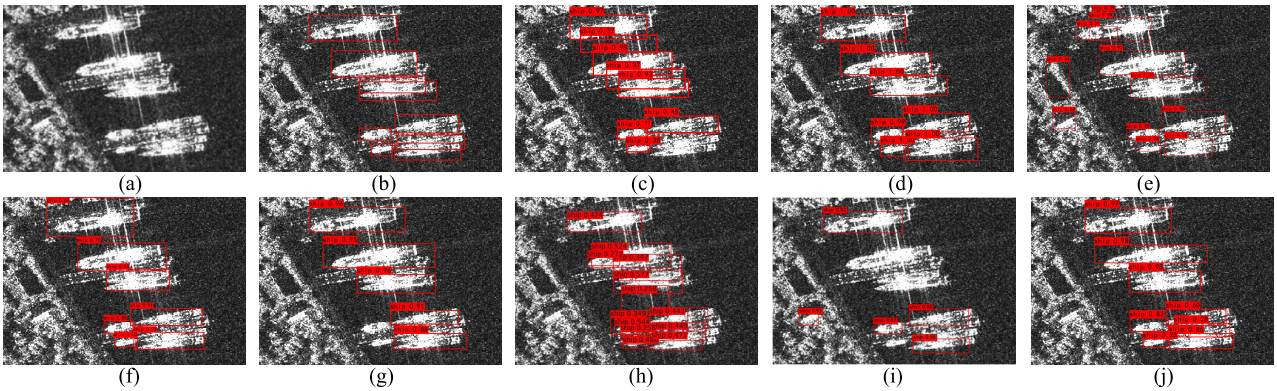


Fig. 15. Detection and recognition results of SAR images with nine ships in the background of the inshore region. (a) Original image. (b) Accurate target position. (c) SSD. (d) YOLOv3. (e) Faster-RCNN. (f) YOLOv4. (g) GoogLeNet. (h) FCOS. (i) DETR. (j) Proposed method.

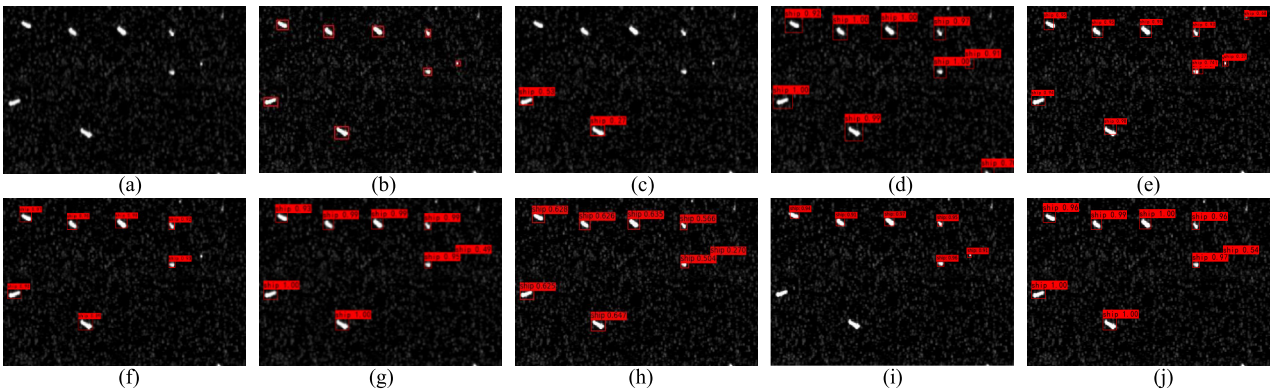


Fig. 16. Detection and recognition results of SAR images with eight ships in the background of the open sea. (a) Original image. (b) Accurate target position. (c) SSD. (d) YOLOv3. (e) Faster-RCNN. (f) YOLOv4. (g) GoogLeNet. (h) FCOS. (i) DETR. (j) Proposed method.

Fig. 14. A total of three targets are detected by SSD and Faster-RCNN, which include a real ship target and two nonship targets. YOLOv3 does not detect the ship target. A target detected by YOLOv4, but it is a nonship target. YOLOV3 and YOLOV4 are one-stage detection methods. It directly produces the category probability and position coordinate value of the object. After a single detection, the final test result can be obtained directly. Therefore, in a complex background, YOLOV3 and YOLOV4 are likely to cause the ship target not to be detected, or the fake ships targets to be detected. A total of two targets are detected by GoogLeNet and FCOS, which are a real ship target and a nonship target. The real target of the ship is correctly detected and recognized by the proposed method and DETR. The recognition rate of the proposed method is 0.5 higher than that of DETR. There are nine real ship targets in Fig. 15. A total of eight targets are detected by SSD and YOLOv3, which include seven real ship targets and a nonship target. However, two ship targets are missed by SSD and YOLOv3. A total of 11 targets are detected by Faster-RCNN, which include seven real ship targets, three nonship targets, and a repeatedly detected ship target. A total of seven targets are detected by YOLOv4, but two ship targets are missed. A total of five targets are detected and identified by GoogLeNet, but four ship targets are missed. A total of 14 targets are detected by FCOS, which include nine real ship targets, a nonship targets, and four repeatedly detected ship targets. The reason is that the anchor frame of FCOS is designed as anchor-free, which is not effective for overlapping target detection, resulting in redundant bounding boxes for overlapping ship targets in complex backgrounds. A total of five targets are detected by DETR, which include four real ship targets and a nonship targets. Multiscale feature detection is not used for DETR, resulting in poor DETR performance in the target detection of ships in complex background. There are eight targets correctly detected and recognized by the proposed method. A ship target is missed by the proposed method. From the detection results in Figs. 13–15, the proposed method has a good detection preference for ships of different sizes in various backgrounds. The main network of the proposed method extracts rich feature information through APNB and ISEB. Then, the proposed method uses FPN's ideological fusion of the feature information in different positions in the main network for further prediction. Therefore, the proposed method can not only effectively distinguish the interference of some strong scatterers on the shore and the sea but also improve the accuracy of ship target detection and recognition.

Fig. 16 shows the ship target under the background of the open sea. The characteristic of this type of target is the simple background environment. However, if the ship target is relatively small, missed detection is easy to occur. The false alarm will increase when there is strong scattering at the sea surface, such as whitecaps and sea spikes. There are eight real ship targets in Fig. 16. Two targets are detected and recognized by SSD, but six ship targets are missed. The reason is that the low-level characteristic convolutional layers of SDD are small, and the feature extraction is insufficient,

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TABLE XI
SSDD SHIP NUMBER DETECTION RESULTS

Method	GT	TP	FN	FP
SSD	262	112	150	11
YOLOv3	262	230	32	12
Faster-RCNN	262	227	35	24
YOLOv4	262	129	133	28
GoogLeNet	262	226	36	30
FCOS	262	234	28	18
DETR	262	221	41	22
Proposed method	262	236	26	7

TABLE XII
EVALUATION INDICATIONS OF SSDD DETECTION RESULTS

Method	MAP (%)	Precision (%)	Recall (%)	F1 (%)
SSD	79.53	91.06	42.75	58
YOLOv3	93.87	95.15	82.44	88
Faster-RCNN	89.57	90.38	89.69	90
YOLOv4	88.25	90.45	75.95	83
GoogLeNet	86.72	88.28	86.26	87
FCOS	90.47	92.51	89.21	91
DETR	90.56	91.24	84.26	88
Proposed method	96.42	97.51	90.08	93

resulting in a large number of missed inspections. Among the nine targets detected and recognized by YOLOv3, eight targets are real ship targets and one target is a nonship target. A total of 12 targets are detected and recognized by Faster-RCNN, which include eight real ship targets, a nonship target, and three repeatedly detected ship targets. Seven targets are detected and recognized by YOLOv4, but a ship target is missed. Three targets are detected and recognized by DETR, but five ship targets are missed. The current existing network detectors usually have multiscale features, and small object targets are usually detected on high-resolution feature charts. However, DETR does not use multiscale features to detect, resulting in the missing of small ship targets. Eight targets are correctly detected and recognized by GoogLeNet, FCOS, and the proposed method. The proposed method has a higher rate of ship recognition than GoogLeNet and FCOS. From the detection results, the proposed method can detect ships and accurately locate ships for SAR images that are severely interfered with by sea clutter and have a relatively indistinct background. Therefore, the proposed method has a good detection ability for the target in this scene and a certain anti-interference ability.

The detection results of the number of SSDD ships are shown in Table XI. In Table XI, TP is the number of true positives, FN is the number of false negatives, FP is the number of false positives, and GT is the ground truth. From Table XI, in the testing set of 116 images and 262 ship targets, the proposed method correctly detected 236 targets, missed 26 targets, and falsely detected seven targets. Compared with SSD, YOLOv3, Faster-RCNN, YOLOv4, GoogLeNet, FCOS, and DETR, the number of targets correctly detected by the proposed method is increased by 124, 6, 9, 107, 10, 2, and 8, respectively. The number of false alarms is reduced by 4, 5, 17, 21, and 23, respectively.

The evaluation indicators of SSDD detection results are shown in Table XII. In Table XII, from the comprehensive performance evaluation index, the MAP of the proposed method reaches 96.42%. It is 16.89%, 2.55%, 6.85%, 8.17%, 9.7%, 5.95%, and 5.86% higher than SSD, YOLOv3, Faster-RCNN, YOLOv4, GoogLeNet, FCOS, and DETR, respectively. The F1 value of the proposed method reaches 93%, which is significantly better than other methods. From the unilateral evaluation index, the Precision and Recall of the proposed method reach 97.51% and 90.08%, which are 6.45% and 47.3% higher than SSD. Too many ship targets are missed by SSD. The reason is that low-level features are used to detect small targets, and the number of low-level feature convolution layers is small, so there is a problem of insufficient feature extraction. Compared with Faster-RCNN, it increased by 7.13% and 0.39%, respectively. Whether the feature extraction network of Faster-RCNN is VGG or ResNet, its feature map is only a single layer, and the resolution is usually small. These are not conducive to small objects and multiscale object detection. Compared with YOLOv3, it increased by 2.36% and 7.64%, respectively. Compared with YOLOv4, it increased by 7.06% and 14.13%, respectively. Compared with GoogLeNet, it increased by 9.23% and 3.82%, respectively. YOLOv3 and YOLOv4 use COCO dataset to cluster the default anchors, but they are not always suitable for SAR image ship target detection, resulting in a low recall rate and poor precision. Compared with FCOS, it increased by 5% and 0.87%, respectively. FCOS is not good for overlapping object detection of similar size. Compared with DETR, it increased by 6.27% and 5.46%, respectively. Multiscale feature detection is not used in DETR, resulting in poor performance of DETR in small object detection. From Tables XI and XII, the proposed method has achieved better detection results than other methods.

The experimental results show that the proposed method has more advantages over other target detection and recognition methods. Compared with SSD, YOLOv3, Faster-RCNN, YOLOv4, GoogLeNet, FCOS, and DETR, MAP, F1, Precision, and Recall of the proposed method for SAR image target detection and recognition are improved and false targets are greatly reduced.

V. CONCLUSION

The method of NLCANet is proposed in this article. First, we propose to select GoogLeNet as the basic framework of the proposed method, which not only increases the width and depth of the network and improves the parameter utilization but also alleviates the problem of overfitting. Second, we propose to perform a simple convolution on the input SAR image and to input it to APNB for non-local operations. The purpose is to make the extracted features no longer limited to adjacent points and extract more detailed feature information. Finally, we propose to input the feature information extracted by APNB to Inception and SEB (ISEB) to obtain the channel dependencies established based on feature fusion

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of different scales and, at the same time, alleviate the problem of overfitting caused by APNB. Experiments on the MSTAR dataset show that the proposed method outperforms other CNN classification and the state-of-the-art machine learning methods. Meanwhile, the SSDD is used in this article to achieve SAR multitarget detection and recognition in complex scenes. The experimental results show that compared with YOLOv3 and YOLOv4, the detection ability of small targets and the distinguishing performance in complex backgrounds are greatly improved, and better recognition results are achieved. NLCANet improves the accuracy of SAR image target recognition and proves the effectiveness of using APNB and SEB for SAR image target recognition. There are still shortcomings in the proposed method. Compared with the state-of-the-art machine learning methods, the testing time of NLCANet is too long. Compared with the DL method, the network structure of the NLCANet needs to be optimized further to reduce the number of parameters while ensuring the recognition accuracy.

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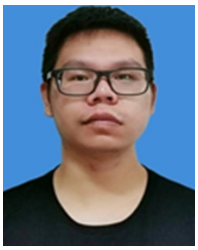
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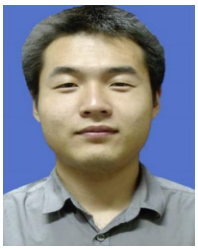
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